

Deployable Intrusion Detection for the CAN Bus: An Event-Level Evaluation Protocol and Identifier-Agnostic Detectors that Transfer Across Vehicles

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Abstract

Learning-based intrusion detection for the Controller Area Network (CAN) is routinely reported above 99% accuracy, yet two properties that decide whether a detector can be fitted to a vehicle are rarely measured: how many false alarms it raises per hour of ordinary driving, and whether it still works on a vehicle it was not trained on. We address both with an event-level evaluation protocol – capture-level cross-validation, whole held-out normal drives for calibration, k -of- n alarm smoothing, a cap on the fraction of time the alarm is active, and detection reported at a stated false-alarm budget – and with a detector designed to satisfy it. Applying the protocol to three public corpora yields four results. First, the most widely used benchmark encodes its own answer: in Car-Hacking every attack is separated from the next by a gap of exactly 3.0 s (coefficient of variation 0.001–0.003), so a logistic model whose only feature is the time since the last attack attains a pre-onset AUC of 1.000 where a spatio-temporal graph network attains 0.851, and window-level detection is saturated at AUC 1.000. Second, on the ROAD corpus a 98 k-parameter convolutional-recurrent network matches a 1.05 M-parameter graph model ($p = 0.81$ over five seeds, identical detected-event sets), both reaching the same 27-of-29 ceiling, so architectural complexity does not pay. Third, five evaluation choices – threshold-grid resolution, seed count, the alarm duty cap during threshold selection and again in reported metrics, and whether held-out normal driving comes from different drives – each individually reverse the apparent ranking of these models. Fourth, on can-train-and-test the same detectors transfer to nothing (0 of 44 attack events over four vehicle pairs, three of them cross-manufacturer) and unlabelled target-vehicle traffic does not repair this, whereas representations that

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describe every frame relative to its own arbitration identifier recover 69–78% of those events with no labels from the target vehicle. Two such identifier-agnostic detectors with OR-ed alarms detect 81% of unknown-vehicle events at 0.3/h false alarms using two 98 k-parameter models (0.20 MB int8, $\approx 40\%$ of one CPU core). Code, configurations and per-event results are released.

Keywords: Controller Area Network, intrusion detection, evaluation methodology, cross-vehicle generalisation, automotive security, false-alarm budget, identifier-agnostic features

1. Introduction

The Controller Area Network (CAN) has been the backbone of in-vehicle communication for four decades and still carries brake, steering, transmission and engine traffic in essentially every production vehicle. It was designed for reliability in an electrically hostile environment, not for security: frames are broadcast, carry no sender identity, and are neither authenticated nor encrypted. Any component with bus access – an infotainment head unit, a telematics modem, an aftermarket dongle, or a physically tapped wire – can therefore inject frames that other electronic control units (ECUs) will act upon. Because retrofitting cryptographic authentication to a deployed fleet is slow and constrained by frame size and ECU budgets, intrusion detection has become the pragmatic line of defence, and a large literature now applies machine learning to the problem (Rajapaksha et al., 2023b; Ghadi et al., 2026).

That literature reports remarkably uniform success. Convolutional, recurrent, attention-based and hybrid detectors report accuracies and F_1 scores between 0.99 and 1.00 on the public benchmarks (Song et al., 2020; Lo et al., 2022; Rai et al., 2025; Saravanan et al., 2025; Zhao et al., 2025), and recent lightweight designs report the same accuracy with two orders of magnitude fewer parameters (Le et al., 2026b; Singh et al., 2026). Taken at face value, in-vehicle intrusion detection would appear to be solved, with only engineering left to do.

Two questions that a vehicle integrator must ask are, however, almost never answered. The first is the false-alarm rate in operational terms. A detector that scores $F_1 = 0.99$ on 100 ms windows can still raise hundreds of alarms per hour of ordinary driving, because windows are numerous and the base rate of attacks is essentially zero; the base-rate problem is well known in network intrusion detection (Flood et al., 2024; Goldschmidt et al., 2025) but is rarely translated into automotive reporting practice. In our own

measurements, thresholds tuned for window- F_1 produced between 159 and 6269 false alarms per hour – numbers that no vehicle programme would accept, and which the corresponding F_1 scores conceal entirely. The second question is transfer. Fleets contain many vehicle models, each with its own arbitration identifiers, message schedules and payload encodings; a detector that must be retrained, with labelled attacks, for every model is a very different product from one that can be shipped once. The recently released can-train-and-test corpus (Lampe and Meng, 2024) was built precisely to expose this, and a companion analysis (Kidmose and Meng, 2025) shows it reveals vehicle overfitting that the older corpora cannot.

This paper asks both questions and answers them empirically. We deliberately keep the detectors ordinary – a convolutional-recurrent network over raw frames, a spatio-temporal graph-attention network over arbitration identifiers, gradient-boosted trees over window features, and two unsupervised statistical detectors – and put the effort into the evaluation and into the representation that is fed to them. Every result in this paper is reported at the level of *events*: an alarm is a rising edge of a k -of- n rule over window scores, a false alarm is such an edge on attack-free traffic, a detection is an alarm that fires inside an attack episode, and latency is measured from the onset of that episode. Detection is always reported at a stated false-alarm budget, together with a cap on the fraction of time the alarm is active, so that a detector which simply never clears cannot appear to be excellent.

Applying that protocol produces a sequence of findings that runs against the prevailing narrative. On Car-Hacking, the benchmark itself supplies the answer: its attacks follow a fixed timetable, so “forecasting” an attack reduces to counting seconds, and a one-feature timer beats a spatio-temporal graph network at every horizon we tested. On ROAD (Verma et al., 2024), whose attacks are physically verified and far stealthier, a 98 k-parameter model matches a 1.05 M-parameter one, both stopping at the same ceiling imposed by the data rather than by the model. On can-train-and-test, identifier-embedding detectors – the standard design – detect *nothing* at a usable false-alarm rate on a vehicle they were not trained on, and no quantity of unlabelled traffic from that vehicle repairs them. The remedy is a change of representation rather than of architecture: describing each frame only relative to its own arbitration identifier recovers most of the lost detection, and running two such representations as separate detectors whose alarms are OR-ed recovers more still, at a cost of 0.20 MB and roughly 40% of one CPU core.

We also document, as first-class results, five evaluation choices that each reversed the apparent ranking of our own models during this study. Coarse

threshold grids, three seeds instead of five, the absence of an alarm duty cap at selection time or at reporting time, and calibrating on time slices of the same drives used for testing each produced a confident, publishable-looking improvement that disappeared under a stricter protocol. We report them because they are cheap to avoid once named, and because they suggest that part of the uniform success in the literature may be protocol rather than progress – a concern voiced independently by Le et al. (2026a), who find that strict de-duplication of a popular corpus collapses deep-model performance, and by Koltai et al. (2026) in a cross-dataset study.

Contributions... This paper contributes:

1. an event-level evaluation protocol for CAN intrusion detection with an explicit false-alarm budget, alarm duty cap, capture-level cross-validation and whole held-out normal drives (Section 5, Algorithm 2);
2. evidence that the standard benchmark rewards learning the injection schedule, together with two dataset artefacts that affect anyone using it (Section 9);
3. a controlled demonstration of five evaluation choices that each reverse model rankings (Section 10);
4. a cross-vehicle study over four vehicle pairs showing that identifier-embedding detectors do not transfer, that unlabelled adaptation does not fix them, and that identifier-agnostic representations do transfer (Section 11);
5. a paired detector combining two identifier-agnostic representations, with an ECU cost budget (Sections 12 and 14); and
6. a public, scripted replication package (Section 16).

Organisation... Section 2 reviews datasets, detectors and evaluation practice, and positions this work. Section 3 states the threat model. Section 4 describes the system architecture. Section 5 gives the formal model of windows, features, alarms and the budget-constrained detection metric, and Section 6 the algorithms. Section 7 describes the three corpora and the artefacts we found, and Section 8 the experimental setup. Section 9 reports what the benchmarks measure, Section 10 the evaluation traps, Section 11 the cross-vehicle study, Section 12 the paired detector, Section 13 the ablations and Section 14 deployment cost. Section 15 discusses implications and threats to validity, Section 16 reproducibility, and Section 17 concludes.

2. Related work

2.1. Attacks on the CAN bus

The attacks studied in this literature follow the taxonomy set out by Verma et al. (2024). *Fabrication* attacks add frames: either indiscriminately, as in a denial-of-service flood of high-priority identifiers, or targeted at one identifier so that the spoofed value competes with the legitimate one. Because the receiving ECU generally acts on the most recent frame, a targeted fabrication attack succeeds if the attacker transmits often enough to win the race, which doubles the observed rate of the victim identifier – a side effect that most published detectors exploit, whether or not they say so. *Suspension* attacks silence a legitimate ECU, removing frames rather than adding them, and leave no injected frame to label. *Masquerade* attacks combine the two: the attacker suppresses the victim ECU and transmits in its place, so the frame rate, the identifier set and the inter-arrival pattern all remain nominal and only the payload semantics are wrong. Masquerade attacks are the practical concern for a capable adversary and the hardest case for any detector, which is why we report them separately throughout this paper.

The feasibility of these attacks on production vehicles was established by remote-compromise demonstrations (Miller and Valasek, 2015), and the difficulty of attributing a frame to a sender – CAN carries no source address – motivated physical-layer defences such as clock-skew fingerprinting of transmitting ECUs (Cho and Shin, 2016). Physical-layer approaches are complementary to the traffic-analysis detectors considered here but require access to signal-level measurements that a software monitor on a gateway ECU does not have.

2.2. Public CAN intrusion detection corpora

Table 1 summarises the corpora that dominate the field and the properties that matter for this paper. The HCRL Car-Hacking corpus (Song et al., 2020) provides four message-injection attacks recorded on one vehicle and is the most frequently used benchmark; the companion Survival Analysis corpus (Han et al., 2018) adds two further vehicles. Verma et al. (2024) survey the available data and argue that public corpora contain attacks that are “too easily detectable”, then release ROAD: physically verified fabrication, fuzzing and accelerator attacks plus simulated masquerade variants on a dynamometer-mounted vehicle, with a comprehensive metadata description. Lampe and Meng (2024) release can-train-and-test, four vehicles from two manufacturers with equivalent attack captures and, crucially, predefined known/unknown vehicle subsets; Kidmose and Meng (2025) pit

sixteen machine-learning detectors against six corpora and conclude that can-train-and-test “paints a clearer picture” because it can reveal overfitting to a vehicle type. Rajapaksha et al. (2024) contribute real moving-vehicle attacks with masquerade and suspension variants. Le et al. (2026a) study CICIoV2024 and show that once redundant frames are removed, deep models fail to generalise – evidence that some reported gains reflect data redundancy rather than detection ability.

Table 1: Public CAN corpora used or discussed in this work. “Events” counts labelled injection episodes rather than frames. Properties relevant to our protocol are marked.

Corpus	Veh.	Attack style	Fixed schedule	Masquerade	Vehicle split
Car-Hacking (Song et al., 2020)	1	fabrication	yes (3.0 s gaps)	no	no
Survival Analysis (Han et al., 2018)	3	fabrication	yes	no	partial
ROAD (Verma et al., 2024)	1	fabrication, fuzzing, targeted	no	sim.	no
CAN-MIRGU (Rajapaksha et al., 2024)	1	real injection, simulated	no	yes	no
can-train-and-test (Lampe and Meng, 2024)	4	nine attack types	no	yes	yes

2.3. Detector families

Detectors divide broadly into three families. *Sequence models over raw frames* treat the arbitration identifier stream, and sometimes the payload, as a time series or an image: deep convolutional networks (Song et al., 2020), recurrence plots (Desta et al., 2022), hybrid convolutional–recurrent stacks (Lo et al., 2022; Singh et al., 2026) and compact convolutional designs aimed at ECU budgets (Le et al., 2026b). *Statistical and signal-level models* use inter-arrival regularity, payload entropy or decoded signals: survival analysis (Han et al., 2018), n -gram analysis with explicit microcontroller budgets (Stabili et al., 2022), cosine-similarity self-similarity tests (Kwak et al., 2021), autoencoder ensembles that target masquerade attacks (Rajapaksha et al., 2023a), and multimodal combinations of time-interval and signal models (Kang et al., 2024). *Relational models* represent the bus as

a graph over identifiers, arguing that structural relationships carry attack evidence (Khreasat et al., 2026). Optimisation-heavy variants tune such detectors with metaheuristics (Ileri et al., 2025).

The three families differ in the evidence they use, and that determines which attacks they can see at all. Sequence models over identifier streams are sensitive to rate and ordering, so they detect fabrication reliably and masquerade poorly, since masquerade preserves both. Payload and signal-level models can in principle detect masquerade, but only when the substituted value is itself unusual; when the attacker writes a value the victim identifier does emit at other times, detection requires comparing identifiers with one another. Relational models address exactly that comparison. Our tier analysis in Section 9 makes this dependency explicit and measures it, rather than leaving it implicit in an aggregate score.

2.4. Transfer, adaptation and federated learning

The dependence of a detector on the vehicle it was trained on has been recognised, but mostly addressed indirectly. Khatri et al. (2023) apply transfer learning so that a detector trained on one attack corpus can be fine-tuned for new attack patterns with less training time, which addresses transfer across *attacks* rather than across vehicles. Althunayyan et al. (2024) propose hierarchical federated learning so that detectors can be trained across a fleet without centralising data, and report improvements when the participating vehicles see diverse driving conditions; the federated setting assumes, however, that each participant contributes labelled or at least representative data, which does not solve the problem of a vehicle model that has never been instrumented for attacks. Kidmose and Meng (2025) show empirically that a detector’s apparent quality depends on the corpus it is measured on and that can-train-and-test exposes vehicle overfitting; their analysis stops short of measuring what happens to the decision threshold under a false-alarm budget, which is the aspect we find decisive.

The practical question – can a detector trained on vehicle A be fitted to vehicle B using only unlabelled traffic from B – is, to our knowledge, unanswered in the literature. We answer it negatively for identifier-embedding detectors (Section 11), which is a useful negative result because that design predominates, and positively for identifier-agnostic representations.

2.5. Representations closest to this work

Two recent works are closest to ours in spirit. Khreasat et al. (2026) stress-test statistical, structural-transition and graph-topology features under cross-attack and cross-dataset shift, and report a hierarchy of what trans-

fers: structural transition features are robust within a platform, while attacks that perturb only payload semantics remain outside their scope. Hegde et al. (2026) propose per-identifier behavioural residualisation – window statistics z-scored against each identifier’s own baseline – and argue that the representation, not the detector, drives performance; they quantify two coverage boundaries (novel-identifier flooding and cross-identifier fuzzing). Neither work evaluates cross-*vehicle* transfer under a false-alarm budget, which is the gap this paper fills; we re-implement the residualisation representation and evaluate it side by side with ours under one protocol.

2.6. Evaluation practice

Concerns about benchmark-driven conclusions are long-standing in network intrusion detection (Flood et al., 2024; Goldschmidt et al., 2025) and are now surfacing in the automotive setting. Sharmin et al. (2024) survey benchmarking frameworks and propose an evaluation design space spanning detector type, attack model, workload and metrics. Koltai et al. (2026) build a cross-dataset benchmark and show that measured performance depends strongly on dataset idiosyncrasies. Ghadi et al. (2026) review deployment claims specifically, introducing explicit criteria for calling a detector “lightweight” or “real-time” and noting that such claims are frequently made without substantiation. Table 2 compares the reporting practice of representative recent work with the protocol used here.

Three observations follow from Table 2. First, false-alarm rates expressed per unit of driving time are almost absent, even though that is the quantity a vehicle programme must budget for; papers instead report false-positive *rates* per window, which are not comparable across window sizes and which hide the absolute alarm count. Second, credit is almost always assigned per window rather than per attack episode, so a detector that fires once in the middle of a 20 s attack and a detector that fires continuously score very differently in F_1 while being operationally similar, and conversely a detector that is permanently in alarm can score well on both. Third, only two of the surveyed works evaluate on a vehicle absent from training, and none reports what happens to the decision threshold when the vehicle changes – which, as Section 11 shows, is where the problem actually lies. Our protocol addresses the three gaps directly rather than proposing another architecture.

2.7. Research gap

Three gaps emerge from this review, and they are connected. The first is *operational reporting*: detection quality is almost always expressed per window, while the quantity a vehicle programme must budget for – alarms

Table 2: Reporting practice in representative recent CAN IDS work. ✓ = reported, ◦ = partially reported, – = not reported. “FA/h” = false alarms per hour of normal driving; “Event” = detection credited per attack episode rather than per window; “Unseen veh.” = evaluated on a vehicle absent from training; “Cost” = measured inference cost on CPU or ECU-like hardware. Entries reflect what each paper reports, not the quality of its method.

Work	Corpora	FA per h	Event level	Unseen vehicle	Cost	Headline metric
Song et al. (2020)	Car-Hacking	–	–	–	–	F_1 , FNR
Desta et al. (2022)	Car-Hacking, own	–	–	–	–	accuracy
Lo et al. (2022)	Car-Hacking	–	–	–	–	accuracy, FAR
Stabili et al. (2022)	own	–	–	–	✓	detection vs memory
Rajapaksha et al. (2023a)	two public	–	–	–	◦	F_1 , latency
Kang et al. (2024)	X-CANIDS, SynCAN	–	–	–	–	TPR, TNR
Ileri et al. (2025)	three public	–	–	–	–	accuracy
Le et al. (2026b)	four public	–	–	◦	✓	accuracy, ms, MB
Singh et al. (2026)	Car-Hacking, ROAD	–	–	–	◦	F_1 , parameters
Khreasat et al. (2026)	HCRL, ROAD	–	–	◦	–	ROC-AUC transfer
Hegde et al. (2026)	HCRL, ROAD	–	–	–	–	F_1 , ROC-AUC
Kidmose and Meng (2025)	six public	–	–	✓	–	F_1 spread
Koltai et al. (2026)	seven public	–	–	◦	–	dataset spread
This work	Car-Hacking, ROAD, can- train-and-test	✓	✓	✓	✓	events at a budget

per hour of driving – is neither reported nor derivable from published numbers, because it depends on window size, alarm smoothing and the duration of the attack-free traffic used. The second is *benchmark saturation*: on the corpora most frequently used, published scores cluster above 0.99, so newly proposed detectors cannot be distinguished from each other or from much simpler ones, and improvements in the third decimal place may reflect protocol differences rather than capability. The third is *vehicle transfer*: only a minority of studies evaluate on a vehicle absent from training, and none measures what a change of vehicle does to the decision threshold, even though a fleet operator cannot label attacks on every model.

These gaps compound. Saturation removes the incentive to report harder quantities; per-window reporting hides the false-alarm cost that saturation conceals; and without an operational metric, cross-vehicle degradation is invisible, because a detector whose alarm never clears can still show a high per-window recall. The protocol in Section 5 closes the three together by scoring events at a stated alarm budget with a duty constraint, on corpora and splits that include a vehicle the detector has never seen.

2.8. Positioning

This work is a measurement and methodology contribution with a constructive component. We do not claim a new architecture: our experiments show that, under a fair protocol, the architectures already available are indistinguishable on the hardest public corpus. We claim (i) a protocol whose numbers survive scrutiny, (ii) a set of measured findings about what current benchmarks and detectors actually deliver, and (iii) evidence that vehicle transfer is a representation problem with a cheap and practical solution.

3. Threat model

We assume an adversary with the ability to transmit arbitrary frames on the CAN bus of a target vehicle, obtained through a compromised connected ECU, a diagnostic port device, or physical access; this is the standard assumption in the corpora we use. The adversary may (i) flood the bus or a single identifier (*rate* attacks), (ii) inject frames carrying values that the target identifier does not normally transmit (*value* attacks), or (iii) suppress a legitimate ECU and transmit plausible values in its place (*masquerade* attacks, a subset of what we term *contextual* attacks, in which each frame is individually plausible and only the combination with other identifiers reveals the manipulation). We assume the defender can observe all bus traffic at a gateway or monitoring ECU, can record a short interval of attack-free traffic

from the vehicle at commissioning time, and cannot obtain labelled attacks for every vehicle model in the fleet. We do not address suppression-only attacks, in which frames are removed but none are injected, nor attacks on higher-layer services above CAN.

4. System architecture

Figure 1 shows the complete pipeline, which has four stages and one calibration path.

Stage 1: framing and references.. Bus traffic is a stream of frames, each carrying a timestamp, an arbitration identifier, a data-length code and up to eight payload bytes. The stream is cut into sliding windows of $F = 64$ consecutive frames with stride 16, and independently into fixed 100 ms time windows that carry the labels and the alarm logic. From the first $W = 30$ s of each capture the system estimates, for every arbitration identifier observed, a small set of reference statistics: median inter-arrival time, frame rate, per-byte minimum and maximum, modal DLC, and the mean and standard deviation of the window statistics used by branch B. These references require no labels, which is what makes per-vehicle commissioning practical.

Stage 2: two identifier-agnostic representations.. Branch A computes twelve per-frame features that are all ratios or deviations with respect to the frame’s own identifier: cadence ratio against the reference inter-arrival time, bit- and byte-change fractions against the previous frame of that identifier, excursion beyond the reference byte range, per-identifier and bus-wide rate ratios, and a DLC-change indicator. Branch B computes fourteen window statistics per identifier – temporal, protocol and payload – and z-scores them against the same references, following the residualisation idea of Hegde et al. (2026). In both branches the identifier is used only to group frames *within* a capture; it never enters the model as a learned embedding and no identifier vocabulary is carried between vehicles. This is the property that survives a change of vehicle.

Stage 3: detectors.. Each branch feeds a small convolutional–recurrent network (ConvGRU) of 98 k parameters: two 1-D convolutions over the frame axis, a bidirectional GRU, and a linear head producing one score per window. The two branches are otherwise identical and are trained independently on the source vehicle’s labelled attacks.

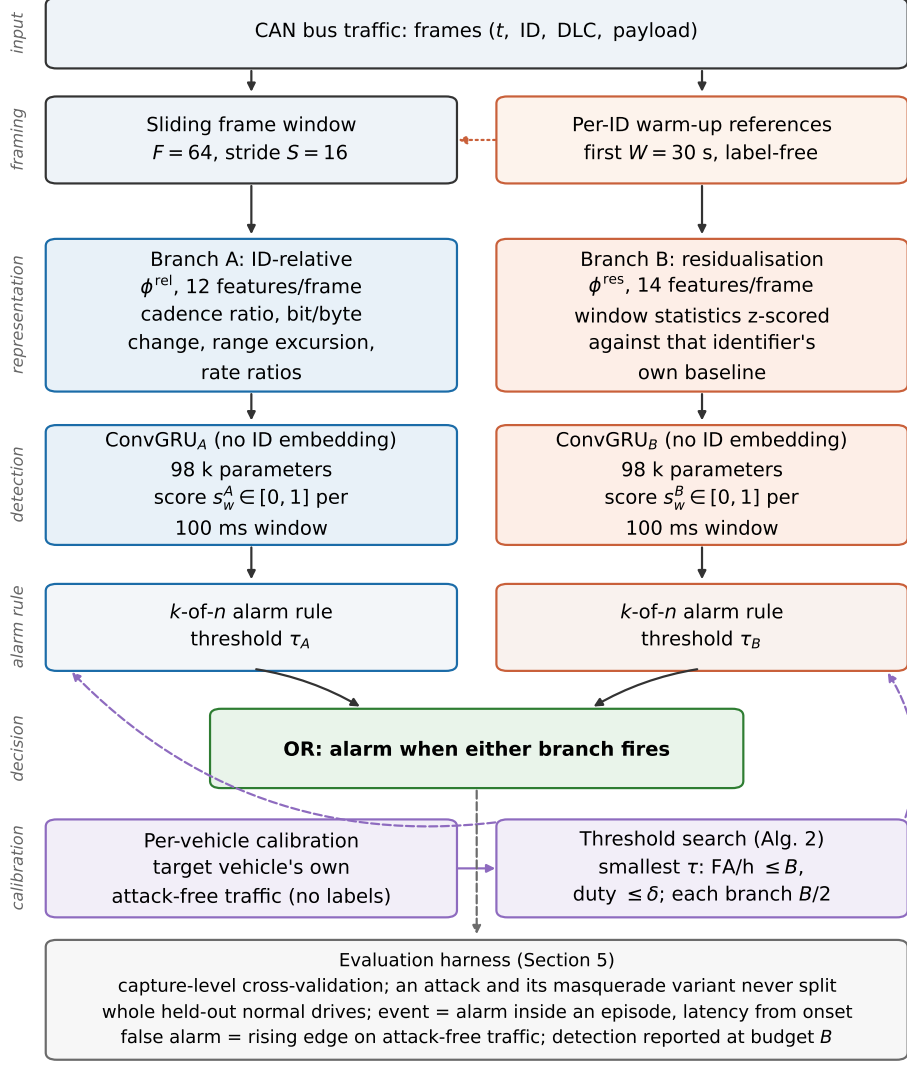


Figure 1: Architecture of the proposed detector and of the evaluation harness that surrounds it. Frames arriving on the bus are cut into sliding windows of F consecutive frames. Per-identifier reference statistics are estimated once from a 30 s warm-up prefix of the capture (orange, label-free). Two identifier-agnostic representations are computed from the same frames: branch A (blue) describes each frame by ratios against its own identifier's references, branch B (orange) by window statistics z-scored against the same references. Each branch feeds an embedding-free ConvGRU that emits one score per 100 ms window; each score stream is converted into alarms by a k -of- n rule whose threshold is calibrated on the target vehicle's own attack-free traffic to satisfy half of the false-alarm budget (purple). An alarm is raised when either branch fires. The grey band lists the evaluation rules of Section 5, which apply to every detector we compare, not only to ours.

Stage 4: alarm logic. Each score stream passes through a k -of- n rule: the branch is in alarm when at least k of the last n window scores exceed its threshold. The system raises an alarm when either branch is in alarm. Because a union of two detectors doubles the opportunity to fire, each branch is calibrated to half of the total false-alarm budget, which keeps the comparison against single branches fair.

Calibration path. Thresholds are not transferred from the source vehicle. At commissioning, attack-free traffic from the target vehicle is scored, and each branch receives the smallest threshold that satisfies both the false-alarm budget and the duty cap (Algorithm 2). No labelled attacks from the target vehicle are required.

5. Mathematical modelling

5.1. Notation and windows

Table 3 lists the notation. Let a capture be a finite frame sequence $\mathcal{F} = \{(t_i, \text{ID}_i, \text{DLC}_i, \mathbf{b}_i)\}_{i=1}^N$ with t_i increasing, $\text{ID}_i \in \mathcal{I}$ the arbitration identifier, and $\mathbf{b}_i \in \{0, \dots, 255\}^8$ the payload. Fix a window length $\Delta = 100$ ms and let $w(i) = \lfloor (t_i - t_1)/\Delta \rfloor$ be the window index of frame i ; window w contains $\mathcal{W}_w = \{i : w(i) = w\}$ and there are T windows.

5.2. Identifier-agnostic representations

Let $p(i)$ be the index of the previous frame with the same identifier. For identifier $u \in \mathcal{I}$ let $\mathcal{W}^u = \{i : \text{ID}_i = u\}$ and let the warm-up set be $\mathcal{W}_W^u = \{i \in \mathcal{W}^u : t_i - t_1 \leq W\}$. The references are

$$m_u = \text{median}\{t_i - t_{p(i)} : i \in \mathcal{W}_W^u\}, \quad r_u = \Delta \frac{|\mathcal{W}_W^u|}{\max(t_{\max} - t_1, 1)}, \quad (1)$$

$$\underline{b}_{u,c} = \min_{i \in \mathcal{W}_W^u} b_{i,c}, \quad \bar{b}_{u,c} = \max_{i \in \mathcal{W}_W^u} b_{i,c},$$

for each byte position $c = 1, \dots, 8$. Branch A maps frame i with $u = \text{ID}_i$ to $\phi^{\text{rel}}(i) \in \mathbb{R}^{12}$. Two of its components are the cadence ratio and the range excursion,

$$\phi_1^{\text{rel}}(i) = \frac{t_i - t_{p(i)}}{m_u}, \quad (2)$$

$$\phi_9^{\text{rel}}(i) = \max_{c \in \{1, \dots, 8\}} \frac{\max(\underline{b}_{u,c} - b_{i,c}, b_{i,c} - \bar{b}_{u,c})}{\max(\bar{b}_{u,c} - \underline{b}_{u,c}, 1)}, \quad (3)$$

Table 3: Notation.

Symbol	Meaning
$\mathcal{F}, N$	frame sequence of a capture and its length
Δ, T, w	window length (100 ms), number of windows, window index
F, S	frame-window length (64) and stride (16)
W	warm-up horizon for per-identifier references (30 s)
$\mathcal{I}$	identifier alphabet of one capture
$m_u, r_u, b_u, \bar{b}_u$	reference cadence, rate and byte range of identifier u
$\phi^{\text{rel}}, \phi^{\text{res}}$	identifier-agnostic representations (12 and 14 features per frame)
$s_w \in [0, 1]$	detector score of window w
$y_w \in \{0, 1\}$	window label: an injected frame is present (after gap filling)
$\mathcal{E} = \{(o_e, c_e)\}$	attack episodes, onset and end window
$a_w(\tau; k, n)$	alarm indicator of the k -of- n rule at threshold τ
B, δ	false-alarm budget (per hour) and alarm duty cap
$\text{FA}(\tau), \rho(\tau)$	false alarms per hour and alarm duty at threshold τ
$D^*(B), \ell_e(\tau)$	events detected at budget B ; detection latency of episode e

the remaining ten being bit- and byte-change fractions with respect to frame $p(i)$, the fraction of bytes below and above the reference range, per-identifier and bus-wide rate ratios, and a DLC-change indicator. Branch B first forms, for each window w and identifier u , a vector $x_{w,u} \in \mathbb{R}^{14}$ of temporal, protocol and payload statistics, then residualises it,

$$\phi_{w,u}^{\text{res}} = \text{clip}\left(\frac{x_{w,u} - \mu_u}{\sigma_u}, -20, 20\right), \quad (4)$$

$$\text{where } (\mu_u, \sigma_u) = (\text{mean}, \text{sd})\{x_{w,u} : w\Delta \leq W\}, \quad (5)$$

and assigns to frame i the vector of its own identifier and window. Both maps are invariant to a relabelling of $\mathcal{I}$: they never use the numeric value or the learned embedding of an identifier, only statistics computed within its own stream. This is the formal sense in which the representation is identifier-agnostic, and it is the property that Section 11 shows to be necessary for cross-vehicle transfer.

5.3. Labels, episodes and alarms

Window w is attacked, $y_w = 1$, if it contains at least one injected frame; because sparse injections leave gaps, we close gaps shorter than 1 s so that

one injection interval yields one episode. Episodes are the maximal runs of $y = 1$, $\mathcal{E} = \{(o_e, c_e)\}_{e=1}^E$, with onset o_e and end c_e .

Given scores s_w and threshold τ , the k -of- n alarm indicator is

$$a_w(\tau; k, n) = \mathbb{K} \left[\sum_{j=\max(1, w-n+1)}^w \mathbb{K}[s_j \geq \tau] \geq k \right]. \quad (6)$$

An episode is *detected* if the rule fires inside it, and its latency is the time from onset to the first such alarm,

$$d_e(\tau) = \mathbb{K}[\exists w \in [o_e, c_e] : a_w(\tau) = 1], \quad (7)$$

$$\ell_e(\tau) = \Delta \cdot \min \{ w - o_e : w \in [o_e, c_e], a_w(\tau) = 1 \}. \quad (8)$$

A *false alarm* is a rising edge on attack-free traffic. Writing $\mathcal{C}$ for the set of attack-free windows (all windows of an attack-free capture, and the pre-onset windows of an attack capture) and $H = \Delta|\mathcal{C}|/3600$ for their duration in hours,

$$\text{FA}(\tau) = \frac{1}{H} \sum_{w \in \mathcal{C}} \mathbb{K}[a_w(\tau) = 1 \wedge a_{w-1}(\tau) = 0], \quad (9)$$

$$\rho(\tau) = \frac{1}{|\mathcal{C}|} \sum_{w \in \mathcal{C}} a_w(\tau). \quad (10)$$

The duty ρ is what prevents a degenerate solution: a threshold so low that the alarm never clears produces exactly one rising edge per capture and therefore an arbitrarily small FA, while being useless. We require $\rho \leq \delta$ with $\delta = 0.02$.

5.4. Budget-constrained detection

The quantity we report throughout is the number of events detected at a false-alarm budget,

$$D^*(B) = \max_{\tau \in \mathcal{T}} \sum_{e=1}^E d_e(\tau) \quad \text{subject to} \quad \text{FA}(\tau) \leq B, \quad \rho(\tau) \leq \delta, \quad (11)$$

where $\mathcal{T}$ is a grid of candidate thresholds. Two properties of (11) matter in practice. First, D^* is not monotone in the grid resolution: a coarse $\mathcal{T}$ can miss the feasible region entirely, which is the first evaluation trap of Section 10; we use $|\mathcal{T}| = 200$ quantiles of the observed scores. Second, when

the threshold must be chosen without access to test data – the deployment case – it is calibrated on separate attack-free traffic $\mathcal{C}_{\text{cal}}$, and the achieved FA on test data is reported alongside, so that a failure of threshold transfer is visible rather than hidden.

For the paired detector, branch scores s^A, s^B with thresholds τ_A, τ_B produce the union alarm

$$a_w^{A \vee B} = a_w^A(\tau_A; k, n) \vee a_w^B(\tau_B; k, n), \quad (12)$$

and each branch is calibrated to budget $B/2$ so that, in the worst case of disjoint false alarms, the union still satisfies B .

6. Proposed algorithms

Algorithm 1 states the label-free construction of per-identifier references and of the two representations; it is executed once per capture and is the only step that touches identifiers. Its cost is linear in the number of frames, $O(N)$ time and $O(|\mathcal{I}|)$ memory for the references, so it is suitable for streaming implementation on a gateway ECU. The warm-up requirement is the practical price: the first W seconds after ignition are used to estimate references and are excluded from evaluation.

Algorithm 1 Identifier-agnostic feature construction (per capture, no labels)

Require: frames $\mathcal{F}$, warm-up W , window length Δ

Ensure: per-frame representations $\phi^{\text{rel}}, \phi^{\text{res}}$

- 1: **for all** identifiers u observed in the warm-up prefix $t_i - t_1 \leq W$ **do**
 - 2: compute references $m_u, r_u, \underline{b}_u, \bar{b}_u$, modal DLC $\triangleright$ Eq. (1)
 - 3: compute window statistics $x_{w,u}$ for warm-up windows and their (μ_u, σ_u)
 - 4: **end for**
 - 5: **for all** frames i with $t_i - t_1 > W$ **do**
 - 6: $u \leftarrow \text{ID}_i$; $p \leftarrow$ index of previous frame of identifier u
 - 7: $\phi^{\text{rel}}(i) \leftarrow$ cadence, change, excursion and rate ratios w.r.t. $(m_u, r_u, \underline{b}_u, \bar{b}_u)$ $\triangleright$ Eq. (2)
 - 8: $\phi^{\text{res}}(i) \leftarrow \text{clip}((x_{w(i),u} - \mu_u)/\sigma_u, -20, 20)$ $\triangleright$ Eq. (4)
 - 9: **end for**
 - 10: **return** $\phi^{\text{rel}}, \phi^{\text{res}}$
-

Algorithm 2 is the calibration procedure. It searches the smallest feasible threshold, which is the most sensitive one that respects both constraints

of (11); the search is over a grid of score quantiles and costs $O(|\mathcal{T}| |\mathcal{C}_{\text{cal}}|)$. The duty test on line 6 is what distinguishes it from the usual practice of maximising F_1 on a validation set, and the fact that it uses only attack-free traffic is what makes per-vehicle commissioning possible without labelled attacks.

Algorithm 2 Per-vehicle threshold calibration under a false-alarm budget

Require: attack-free calibration scores $\{s_w\}_{w \in \mathcal{C}_{\text{cal}}}$, budget B , duty cap δ , rule (k, n)

Ensure: threshold τ^*

```

1:  $\mathcal{T} \leftarrow$  200 quantiles of  $\{s_w\}$ , sorted ascending
2: for all  $\tau \in \mathcal{T}$  do
3:    $a_w \leftarrow$   $k$ -of- $n$  rule applied to  $\{s_w \geq \tau\}$  ▷ Eq. (6)
4:    $\text{FA} \leftarrow$  rising edges of  $a$  per hour;  $\rho \leftarrow$  mean of  $a$  ▷ Eq. (9)
5:   if  $\text{FA} \leq B$  and  $\rho \leq \delta$  then
6:     return  $\tau^* \leftarrow \tau$  ▷ most sensitive feasible threshold
7:   end if
8: end for
9: return  $\tau^* \leftarrow \max \mathcal{T} + \epsilon$  ▷ infeasible: report as such

```

Algorithm 3 is the runtime procedure of the paired detector. Each branch is evaluated independently and the alarms are combined by disjunction, with each branch calibrated to half the budget. The per-window cost is two ConvGRU evaluations plus $O(1)$ alarm bookkeeping; Section 14 measures it.

Algorithm 3 Paired identifier-agnostic detection at run time

Require: frame stream, models g_A, g_B , thresholds τ_A, τ_B from Algorithm 2 with budget $B/2$ each, rule (k, n)

Ensure: alarm events

```

1: collect the warm-up prefix and build references ▷ Algorithm 1
2: for all arriving frame windows  $(i - F, \dots, i)$  with stride  $S$  do
3:    $s^A \leftarrow g_A(\phi^{\text{rel}}(i - F:i)); \quad s^B \leftarrow g_B(\phi^{\text{res}}(i - F:i))$ 
4:   assign  $s^A, s^B$  to the 100 ms window containing  $t_i$  (maximum if several)
5:    $a^A \leftarrow k\text{-of-}n(s^A \geq \tau_A); \quad a^B \leftarrow k\text{-of-}n(s^B \geq \tau_B)$ 
6:   if  $a^A \vee a^B$  and no alarm was active in the previous window then
7:     emit alarm event with timestamp  $t_i$  ▷ Eq. (12)
8:   end if
9: end for

```

7. Datasets

Table 4 summarises the three corpora as we use them, after preprocessing.

Table 4: Corpora as used in this study, after preprocessing. “Normal” is the attack-free traffic available for calibration and false-alarm measurement.

Corpus	Role in this study	Veh.	Events	Normal	Attack types
Car-Hacking (Song et al., 2020)	benchmark critique	1	4×300	481.6 s	DoS, fuzzy, gear, RPM
ROAD (Verma et al., 2024)	detection study	1	29	72 min	fuzzing, targeted injection, masquerade
can-train-and-test (Lampe and Meng, 2024)	cross-vehicle study	4	44 (+26 masq.)	22 min per pair	DoS, spoofing, interval, standstill, masquerade

Car-Hacking. Four attack captures and one attack-free capture from a single vehicle. Parsing the raw logs reveals two artefacts that affect any user of this corpus. First, each attack file contains exactly 300 injection episodes whose inter-episode gaps are 3.0 s with a coefficient of variation of 0.001–0.003 (Fig. 2a); the schedule is therefore almost perfectly periodic. Second, the same 481.6 s attack-free recording (891,068 byte-identical frames) is appended to all four attack files, and `Fuzzy_dataset.csv` additionally contains two recording sessions separated by a 2540 s silence. Without de-duplication the attack-free traffic is counted four times; without splitting at the silence, a chronological split places every attack in the training portion and none in validation or test. We do both.

ROAD. We use the nine dynamometer ambient captures (72 min of attack-free driving) and every attack capture with a labelled injection interval: 15 physically verified injection events, 14 of which also exist as masquerade variants in which conflicting legitimate frames were removed, giving 29 attack events. Frame labels are reconstructed from the published metadata – injection interval, target identifier, and a byte pattern with wildcards. As a consistency check, in non-masquerade captures the target identifier carries exactly twice as many frames inside the interval as there are injected frames (injected plus displaced legitimate), and in the masquerade variants the two

counts coincide; our reconstruction reproduces this for all captures, and every reconstructed episode begins within one window of the published onset. The four accelerator captures carry no injected frames and are excluded.

can-train-and-test.. Four sub-datasets, each nominating a known and an unknown vehicle: Impala→Silverado, Traverse→Forester, Silverado→Forester and Forester→Traverse, the last three crossing manufacturers. We use the attack-free training subset, the attack training subset, the known-vehicle and unknown-vehicle test subsets and the masquerade subset. The target vehicles have up to 98 arbitration identifiers against 52 for the source, which is one reason identifier-embedding models fail to transfer.

8. Experimental setup

Protocol parameters.. $\Delta = 100$ ms, $F = 64$ frames, stride $S = 16$, warm-up $W = 30$ s, gap filling 1 s, duty cap $\delta = 0.02$, alarm rules $(k, n) \in \{(1, 1), (2, 3), (3, 5)\}$ with $(3, 5)$ as default, budgets $B \in \{5, 10, 30, 60\}$ 1/h, threshold grid $|\mathcal{T}| = 200$.

Models.. The ConvGRU has two 1-D convolutions (64 channels, kernel 5, the second strided), a bidirectional GRU (64 units) and a two-layer head; with an identifier embedding of 32 dimensions it has 98 k parameters, and the embedding-free variants used in Section 11 are of the same size. It is trained for 15 epochs of 600 steps, batch 512, AdamW at 10^{-3} , with positive class weight 5. The spatio-temporal graph model (referred to as the graph-attention model) applies two dense graph-attention layers with four heads over the identifier graph of each window, a four-layer Transformer encoder over a 20-window history, per-horizon heads and Monte-Carlo dropout; $d_{\text{model}} = 128$, 1.05 M parameters, AdamW at 3×10^{-4} , 25 epochs with early stopping (patience 6), batch 32. The window GBDT is a histogram gradient-boosting classifier (300 iterations, balanced class weights) over all identifiers’ features in the current window. The two unsupervised detectors are exceedance tests against per-identifier normal ranges, one on frame counts and one on per-(identifier, byte) payload novelty.

Cross-validation.. On ROAD, fold $f \in \{0, 1, 2\}$ tests attack instance $f+1$, validates on the next instance and trains on the remainder; an attack and its masquerade variant always share a split. Ambient captures are assigned whole to training, calibration or test, rotating with the fold. On can-train-and-test we use the corpus’s own known/unknown vehicle split.

Seeds and statistics. Unless stated otherwise, learned models are run with five seeds on ROAD and three on can-train-and-test; we report mean $\pm$ standard deviation. Model pairs are compared with the Mann–Whitney U test over seeds and, for detected-event sets at a common budget, with an exact McNemar test.

Hardware. Training and inference use one NVIDIA RTX 5060 Laptop GPU (8 GB), 16 GB RAM, PyTorch 2.11 with CUDA 12.8. Cost measurements in Section 14 use a single CPU thread at batch size 1.

9. What the standard benchmarks measure

9.1. Car-Hacking rewards learning the schedule

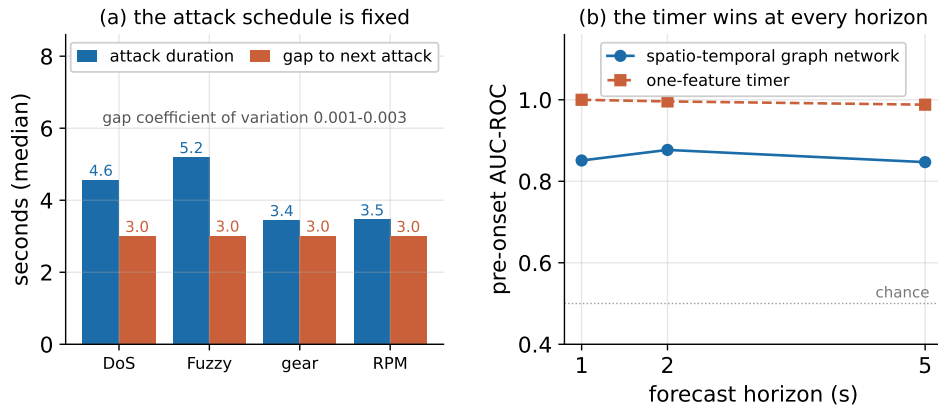


Figure 2: Car-Hacking. (a) Median attack duration and median gap between consecutive attacks for the four attack files; the gap is 3.0 s in every file, with coefficient of variation between 0.001 and 0.003. (b) Pre-onset AUC-ROC – computed only on windows in which no attack is yet in progress – for a spatio-temporal graph network and for a logistic model whose only feature is the time since the last attack. The timer is at or above the network at every horizon, so what is being measured is the injection schedule.

Figure 2a shows why the corpus cannot support claims about anticipating attacks. Its attacks follow a timetable, so the optimal predictor of the next onset is a clock. Figure 2b makes this concrete on the fuzzy capture: restricted to pre-onset windows, a one-feature timer attains AUC 1.000, 0.996 and 0.988 at horizons of 1, 2 and 5 s, against 0.851, 0.877 and 0.847 for the graph network with Monte-Carlo dropout. Pooling all four attack files

the ordering persists at 1 s (0.980 against 0.901). We therefore regard published look-ahead results on this corpus as unsupported unless they report a timing-only baseline.

Detection on the same corpus is saturated: the graph model reaches $F_1 = 0.9997$ at AUC 1.000 when asked whether the current window contains an attack, matching the near-perfect numbers common in the literature (Song et al., 2020; Lo et al., 2022; Rai et al., 2025) and leaving no headroom in which a new method could demonstrate an advantage. This is consistent with the redundancy analysis of Le et al. (2026a) on a different corpus.

9.2. On ROAD, architectural complexity does not pay

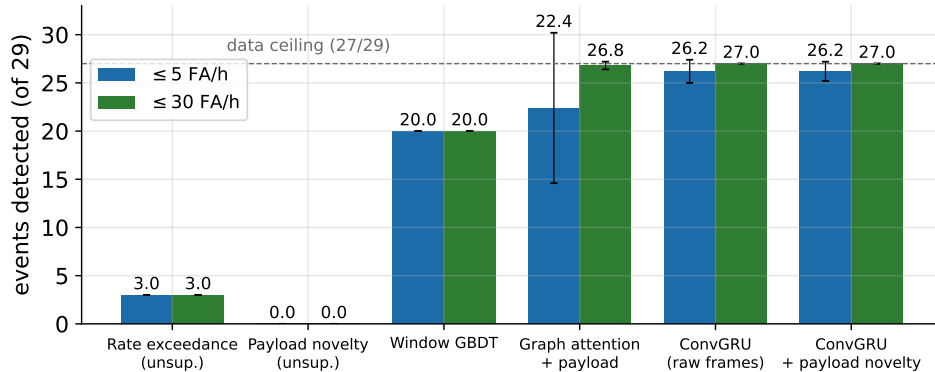


Figure 3: ROAD: events detected of 29 at two false-alarm budgets, 3-of-5 alarm rule, alarm duty $\leq 2\%$. Learned models use five seeds (error bars are one standard deviation); the unsupervised detectors and the GBDT are deterministic given the fold. The dashed line marks the ceiling of 27 events that no configuration exceeds.

Table 5 and Figure 3 report the comparison. The three learned models are statistically indistinguishable: ConvGRU against ConvGRU with payload novelty gives $p = 1.000$ with identical detected-event sets under McNemar, and either against the graph model gives $p = 0.81$ at 5/h and $p = 0.42$ at 30/h under Mann–Whitney. All three stop at 27 of 29 events. The two events that no model detects are the engine-coolant-temperature injection and its masquerade variant, which deliver 43 frames over 4 s; the ceiling is therefore a property of the data, not of the models. The 98 k-parameter model is ten times smaller, twenty-nine times faster per inference (Section 14) and considerably more stable across seeds than the 1.05 M-parameter one.

Figure 4 shows the full trade-off from which those two operating points are taken. The unsupervised detectors are unusable at any budget, the

Table 5: ROAD, 3-fold capture-level cross-validation, 29 attack events, 1.2 h of held-out attack-free driving, 3-of-5 alarm rule. Events detected (mean $\pm$ s.d. over seeds).

Model	Params	$\leq 5/h$	$\leq 30/h$
Rate exceedance (unsupervised)	—	3/29	3/29
Payload novelty (unsupervised)	—	0/29	0/29
Window GBDT (no graph, no history)	—	20/29	20/29
Graph attention + payload novelty	1.05 M	22.4 ± 7.8	26.8 ± 0.4
ConvGRU on raw frames	98 k	26.2 ± 1.2	27.0 ± 0.0
ConvGRU + payload novelty	98 k	26.2 ± 1.0	27.0 ± 0.0

GBDT saturates at 20 events, and the learned sequence models converge to the ceiling once the budget exceeds roughly 20 alarms per hour.

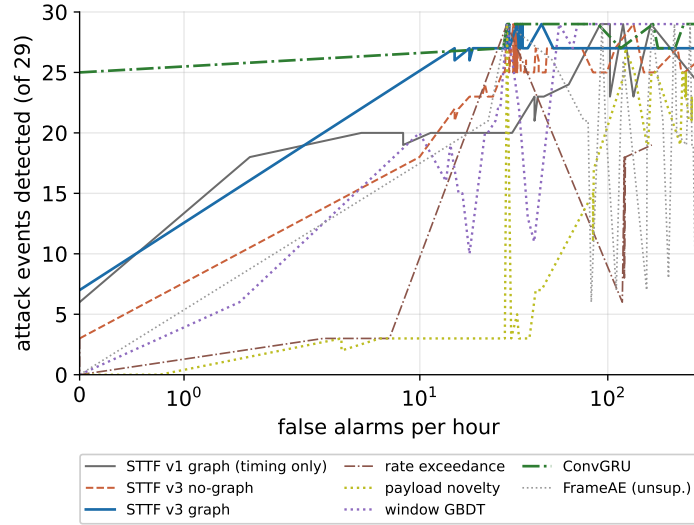


Figure 4: ROAD: attack events detected against false alarms per hour of held-out attack-free driving, sweeping the threshold (3-of-5 rule, duty $\leq 2\%$, folds pooled, seed 0). Curves that rise only at high alarm rates are unusable in a vehicle regardless of their AUC.

9.3. A detectability hierarchy

Attacks fall into three tiers, and the tier decides what evidence is required. Figure 5b quantifies the distinction on ROAD. *Rate* attacks change

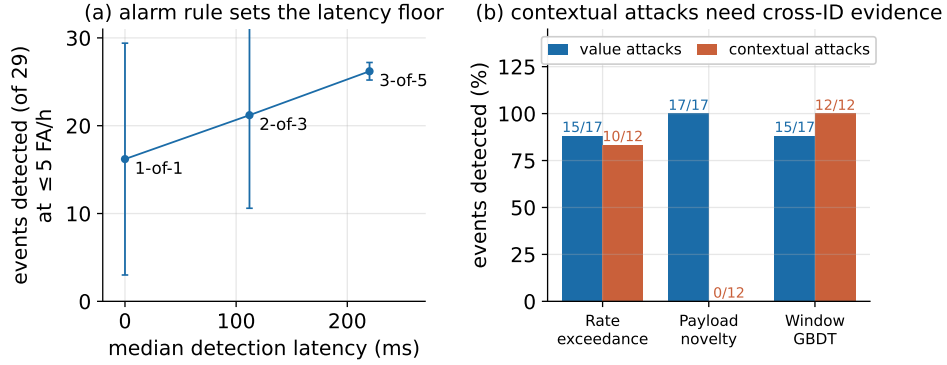


Figure 5: (a) Events detected against median detection latency for three alarm rules (ConvGRU + payload novelty, five seeds, ≤ 5 /h); the rule, not the model, sets the latency floor, and the fastest rule is also the least stable. (b) Detection by attack tier: the per-(identifier, byte) novelty detector finds every value attack and no contextual attack, while a model that sees all identifiers jointly finds all contextual attacks.

an identifier’s frame rate and are found by a frame-count exceedance test (15 of 17 value attacks, 10 of 12 contextual). *Value* attacks write byte values the identifier never normally carries: the per-(identifier, byte) novelty test finds 17 of 17. *Contextual* attacks write an ordinary value that is wrong only given other identifiers – ROAD’s reverse-light attacks set the light state while the vehicle is not in reverse – and the same novelty test finds 0 of 12, while the GBDT, which sees all identifiers jointly, finds 12 of 12. Cross-identifier evidence is therefore necessary for the third tier, but a graph neural network is not: a flat model over all identifiers suffices.

9.4. Latency is set by the alarm rule

Figure 5a and Table 6 separate model latency from rule latency. A k -of- n rule cannot fire before k windows have exceeded the threshold, so the rule imposes a floor of $(k - 1)\Delta$. The unsmoothed 1-of-1 rule detects inside the onset window – a median latency of 0 ms, i.e. below 100 ms – but has no feasible threshold for two of five seeds, whereas the stable 3-of-5 rule costs 220 ms. Reporting a sub-100 ms latency therefore requires stating the alarm rule and its stability; otherwise the number is not comparable between papers.

Table 6: Latency and stability by alarm rule. ConvGRU with payload novelty at $\leq 5/h$, five seeds, ROAD.

Rule	Events detected	Median latency	Note
1-of-1	16.2 ± 13.2	0 ms	no feasible threshold for 2 of 5 seeds
2-of-3	21.2 ± 10.6	112 ms	1 of 5 seeds infeasible
3-of-5	26.2 ± 1.0	220 ms	stable across all seeds

10. Five evaluation traps

Table 7 lists five choices that, applied alone to the same code and data, reverse the ranking of the models in Table 5. We encountered each of them in the course of this study, in the order shown, and each initially produced a confident conclusion that a stricter protocol subsequently removed.

Table 7: Five evaluation choices, each sufficient on its own to reverse model rankings. “Permissive” is the choice that produced the wrong conclusion; “strict” is the setting used in this paper.

Choice	Permissive setting and its conclusion	Strict setting and the true result
Threshold-grid resolution	60 candidate thresholds: payload-novelty features add +4.2 events, $p = 0.029$	200 candidates: $p = 1.000$, identical detected-event sets
Seed count	3 seeds: graph attention adds 11 events over its ablation	5 seeds: $p = 0.81$; single-seed scores range from 7 to 27 events
Duty cap during threshold selection	absent: “29/29 events at 29/h” from an alarm that never clears	present: the alarm must clear; 27/29 at 28/h
Duty cap on reported metrics	absent: 8/8 events on an unseen vehicle	present: the alarm is active 99.9% of the time; 0/8 events
Composition of held-out normal traffic	time slices of the drives used for testing: a 60/h budget appears to transfer	whole held-out drives: the same budget yields 1244–2866/h on unseen drives

The first two are statistical: a coarse threshold grid can miss the feasible region of (11) entirely, and three seeds cannot resolve a difference when the seed standard deviation reaches eight events. The third and fourth concern the degenerate solution that (9) admits without a duty constraint; the fourth is particularly insidious because it appears only when the detector is applied to data unlike its training data, which is exactly the cross-vehicle case. The fifth is a data-composition choice that silently converts a calibration exper-

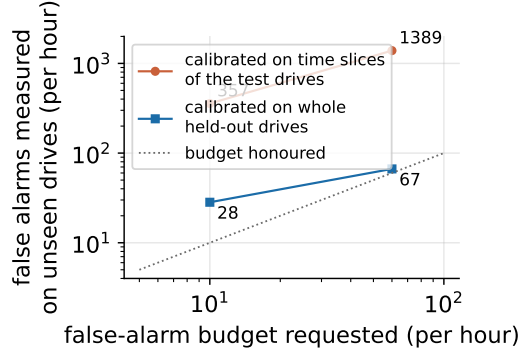


Figure 6: The fifth trap made visible. A threshold is calibrated to a requested false-alarm budget (horizontal axis) and the rate it actually achieves on unseen normal driving is measured (vertical axis); the dotted line is the budget being honoured. Calibrating on time slices of the same drives used for testing (orange) overshoots by one to two orders of magnitude, because the calibration and test traffic are statistically the same drive. Calibrating on whole held-out drives (blue) keeps the achieved rate within a small factor of the request. Graph-attention model, 3-of-5 rule.

iment into an interpolation experiment, and Figure 6 shows its magnitude: a budget of 10/h calibrated on time slices of the test drives yields 357/h in practice, whereas the same budget calibrated on whole held-out drives yields 28/h. None of the five traps is expensive to avoid; all five are invisible in a table of F_1 scores.

11. Cross-vehicle transfer

11.1. Identifier-embedding detectors transfer to nothing

Trained on one vehicle of a can-train-and-test pair and evaluated on the other, the standard identifier-embedding detector detects 0 of 44 events at a valid operating point (Fig. 7, grey). The unconstrained numbers look excellent – with the source vehicle’s threshold it flags 8 of 8 events on the first pair – but its alarm is then active 99.9% of the time, and once the threshold must satisfy a budget on the target vehicle’s own attack-free traffic, detection falls to zero. Removing the payload features changes nothing, so this is not an artefact of one feature set: the learned identifier embeddings and traffic patterns are vehicle-specific, and the target vehicles carry up to 98 identifiers against 52 in the source.

Figure 8 shows that supplying unlabelled traffic from the target vehicle does not help: 0 or 1 of 8 events at a valid operating point, with no

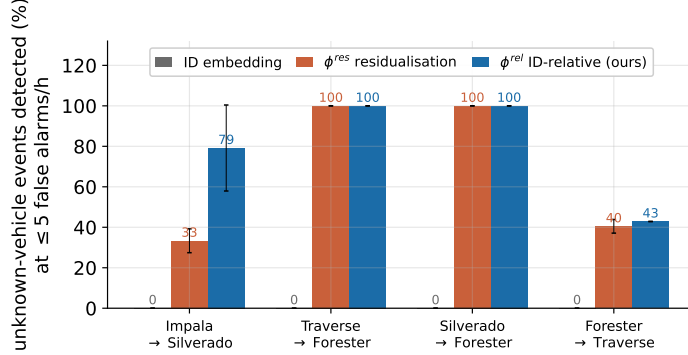


Figure 7: Unknown-vehicle events detected at $\leq 5/h$ false alarms, thresholds calibrated on the target vehicle’s own attack-free traffic without labels, three seeds. The identifier-embedding detector scores zero on every pair; both identifier-agnostic representations transfer.

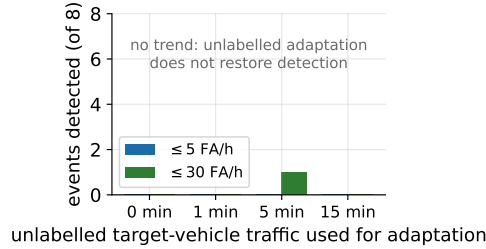


Figure 8: Unlabelled adaptation does not repair transfer. The source-trained detector is given 0, 1, 5 or 15 min of the target vehicle’s attack-free traffic (identifiers added to the vocabulary, frames added as negatives, threshold calibrated on a held-out fifth), and evaluated on target attack captures never used for adaptation.

trend in the amount of traffic. The mechanism is visible in the construction: those identifiers’ embeddings are trained exclusively on normal traffic, so the classifier has never seen what an attack looks like on any of them.

11.2. Identifier-agnostic representations transfer

Table 8 reports the alternative. Both identifier-agnostic representations transfer across all four pairs with no labels from the target vehicle: the per-frame relative representation ϕ^{rel} detects 34.3 of 44 unknown-vehicle events (78%) and the residualisation representation ϕ^{res} 30.3 of 44 (69%), against 0 of 44 for identifier embeddings. Transfer is therefore a property of identifier-agnostic representations in general rather than of one particular feature set,

which also means that the representation of Hegde et al. (2026), published without a cross-vehicle evaluation, transfers.

The two differ in their blind spots, and complementarily so: on masquerade attacks ϕ^{res} reaches 40% against 18% for ϕ^{rel} . Concatenating them into a single 26-feature input fails – unknown-vehicle detection drops to 53% at 5/h and 60% at 30/h, and on one pair the combined model collapses to 0 of 10 because its scores on the target vehicle’s clean traffic saturate near 1.0 so that no threshold satisfies the budget, while with the source vehicle’s threshold it still finds 10 of 10. Discrimination survives; calibration does not. Complementary representations must therefore be combined as separate detectors, which is what Section 12 does.

Table 8: Cross-vehicle detection at $\leq 5/\text{h}$, thresholds calibrated on the target vehicle’s own attack-free traffic, three seeds. Three of the four pairs cross manufacturers.

Vehicle pair	Events	ID embedding	ϕ^{res} residualisation	ϕ^{rel} relative
Impala $\rightarrow$ Silverado	8	0	2.7 ± 0.5	6.3 ± 1.7
Traverse $\rightarrow$ Forester	12	0	12.0 ± 0.0	12.0 ± 0.0
Silverado $\rightarrow$ Forester	10	0	10.0 ± 0.0	10.0 ± 0.0
Forester $\rightarrow$ Traverse	14	0	5.7 ± 0.5	6.0 ± 0.0
Total	44	0 (0%)	30.3 (69%)	34.3 (78%)
Masquerade subsets	26	–	10.3 (40%)	4.7 (18%)

12. Paired identifier-agnostic detectors

Table 9: Paired detector on can-train-and-test, four vehicle pairs $\times$ two seeds. Unknown vehicle: 88 events; masquerade: 52 events. Each branch of the union is calibrated to $B/2$.

Branch	Unknown vehicle (88)		Masquerade (52)		FA/h measured
	$B=5/\text{h}$	$B=30/\text{h}$	$B=5/\text{h}$	$B=30/\text{h}$	
A: ϕ^{rel} relative	59 (67%)	62 (70%)	10 (19%)	16 (31%)	1.2
B: ϕ^{res} residualisation	61 (69%)	63 (72%)	19 (37%)	25 (48%)	1.3
$A \vee B$	71 (81%)	73 (83%)	20 (38%)	27 (52%)	0.3

Because the blind spots are disjoint, we run both representations as separate detectors and raise an alarm when either fires. Table 9 and Figure 9

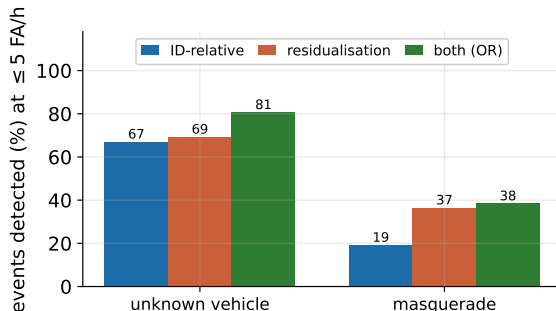


Figure 9: Two identifier-agnostic detectors with OR-ed alarms, each calibrated to half the budget, on the unknown vehicle and on masquerade attacks (four pairs $\times$ two seeds, $\leq 5/h$). The union exceeds both branches on both test sets.

report the result. The union detects 71 of 88 unknown-vehicle events (81%) at 0.3/h measured false alarms, against 67% and 69% for the branches alone at 1.2–1.3/h, and 83% at a 30/h budget. On masquerade attacks it reaches 38% at 5/h and 52% at 30/h, roughly double the relative branch alone. The union improves detection *and* lowers the measured false-alarm rate, because each branch operates at a stricter threshold; it is therefore not a trade-off between the two objectives. Masquerade coverage of 38–52% remains the principal open problem and is consistent with the contextual tier being the hardest case throughout this study.

Operationally the design degrades gracefully. On a vehicle with no labelled attacks, both branches are available because neither requires target labels – only the 30s warm-up and a short attack-free recording for calibration. If labelled attacks for that vehicle later become available, a vehicle-specific branch can be added to the same disjunction without changing the calibration procedure.

13. Ablation study

Table 10 and Figure 10 summarise the ablations; the last row of each block is the configuration we recommend.

On ROAD (upper block), removing the payload-novelty features from the ConvGRU costs nothing at this budget, and adding graph attention with ten times the parameters also changes nothing within noise. Removing the temporal history and the sequence model – the window GBDT, which sees all identifiers of the current window but no past – costs seven events, which locates the useful signal in the short-term temporal structure rather than in

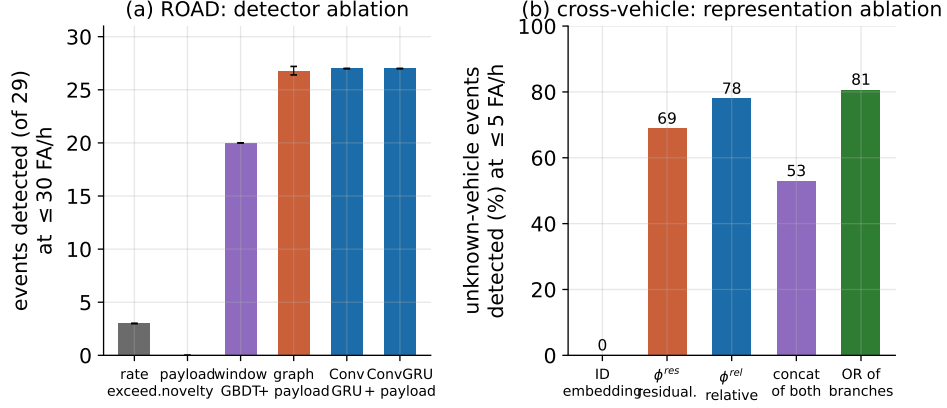


Figure 10: (a) Detector ablation on ROAD at $\leq 30/h$: removing history and the graph (GBDT) costs seven events, while replacing the graph model with a ten-times smaller ConvGRU costs nothing. (b) Representation ablation on can-train-and-test: identifier embeddings transfer to nothing, both identifier-agnostic representations transfer, concatenating them is worse than either, and running them as two detectors is best.

Table 10: Ablations. Upper block: detector components on ROAD (29 events, $\leq 30/h$). Lower block: representation and combination on can-train-and-test (44 unknown-vehicle events, $\leq 5/h$). Δ is the change with respect to the row marked as reference.

Ablation	Configuration	Events	Δ
ROAD detector	ConvGRU + payload novelty (reference)	27.0 ± 0.0	–
	– payload novelty (raw frames only)	27.0 ± 0.0	0.0
	+ graph attention ($10\times$ parameters)	26.8 ± 0.4	–0.2
	– history and sequence model (window GBDT)	20.0	–7.0
	– supervision (payload-novelty exceedance)	0.0	–27.0
Cross-vehicle	ϕ^{rel} relative (reference)	34.3 (78%)	–
	ϕ^{res} residualisation	30.3 (69%)	–4.0
	identifier embedding + payload	0.0 (0%)	–34.3
	concatenation $[\phi^{rel}, \phi^{res}]$	23.3 (53%)	–11.0
	two detectors, OR-ed alarms	81% of 88	+3 pp

the cross-identifier graph. Removing supervision altogether costs everything at a usable budget: the unsupervised novelty test detects no events at 30/h, despite finding every value attack when its threshold is tuned for window- F_1 (Fig. 5b). That contrast is itself a warning about window-level metrics.

Across vehicles (lower block), the representation is decisive. Identifier embeddings lose all 34 events that ϕ^{rel} recovers. Residualisation recovers most of them, confirming that the property that matters is the absence of identifier identity rather than our particular features. Concatenation loses eleven events relative to ϕ^{rel} despite containing strictly more information, because the joint input degrades score calibration on unseen vehicles. Running the two representations as separate detectors recovers three percentage points over the better branch while halving the measured alarm rate.

14. Deployment cost

Table 11: Inference cost, one CPU thread, batch size 1. “Rate” is the number of inferences per second implied by the sliding-window stride; “duty” is the resulting fraction of one core.

Model	Params	int8 (MB)	Median (ms)	p99 (ms)	Rate (s ⁻¹)	Core duty (%)
ConvGRU + payload novelty	98 k	0.10	1.35	1.69	150	20
Graph attention	1.05 M	1.05	38.9	70.1	10	39
Graph attention, 50 MC passes	1.05 M	1.05	1945	3507	10	1945
Paired detector (two ConvGRU)	196 k	0.20	2.70	3.38	150	40

Table 11 reports measured inference cost. The 98 k-parameter detector needs 0.10 MB in int8 and about 20% of one core at the rate implied by its stride; the paired design of Section 12 doubles both. Feature extraction adds 0.3% of a core over real traffic and is negligible. By contrast, the 50-pass Monte-Carlo-dropout uncertainty estimate that accompanies much of the forecasting literature (Gal and Ghahramani, 2016) requires roughly twenty cores at the same window rate, which rules it out for an embedded deployment; five passes, or a deterministic calibration method, would be needed instead. A laptop core is several times faster than a typical automotive Cortex-A or Cortex-R core, so these figures are optimistic lower bounds; the duty column is the quantity that scales with a slower core.

15. Discussion

Benchmark accuracy has stopped being informative.. Two of our corpora are saturated at the window level: on Car-Hacking every model we tried reaches AUC 1.000, and on ROAD every learned model reaches the same 27-of-29 ceiling. Under those conditions, differences between published F_1 scores in the third decimal place carry no information about deployability, while quantities that do – alarms per hour, behaviour on an unseen vehicle, cost on a real core – are rarely reported (Table 2). Our results suggest reallocating effort from architecture to evaluation and representation.

Why complexity does not pay here.. The graph-attention model was designed on the assumption that cross-identifier structure carries attack evidence. Figure 5b shows that the assumption is correct for contextual attacks, but Table 10 shows that a flat model over all identifiers already captures it; what the graph adds beyond that is within seed noise. Meanwhile, the ablation that does matter – removing temporal history – costs seven events. On the evidence here, temporal structure and a sensible representation are the load-bearing components.

The vehicle gap is a representation problem.. The most consequential finding for practice is that a detector which scores well on its own vehicle can detect nothing on another while appearing to work, because the alarm simply never clears. Identifier embeddings are the cause: they tie the model to a vocabulary that does not survive the change of vehicle, and they cannot be repaired with unlabelled traffic because the new identifiers are only ever seen in normal traffic. Once the representation is expressed relative to each identifier’s own behaviour, most of the detection returns. This also implies a practical commissioning procedure: record half an hour of ordinary driving from a new vehicle model, calibrate thresholds on it, and deploy – no labelled attacks required.

What remains hard.. Masquerade attacks are the residual problem: 38–52% coverage with the paired design, and 0 of 12 for a per-identifier novelty test on ROAD’s contextual attacks. They are hard for a principled reason – every frame is individually plausible – and they are exactly the attacks that matter most for a capable adversary. Progress here will require models of the consistency between signals carried by different identifiers, evaluated under the protocol used here rather than at a window- F_1 optimum.

Threats to validity. Our conclusions rest on three public corpora, one of which supplies all cross-vehicle evidence; a fifth vehicle pair could change the numbers, though it is unlikely to restore identifier-embedding transfer given the mechanism. Our re-implementations of a residualisation representation and of frame-sequence baselines follow published descriptions rather than authors’ code, so absolute values may differ from the originals; we therefore draw conclusions only from comparisons run within our own protocol, and we release the code so that the comparison can be repeated. Seed variance is substantial at strict budgets, which is why every learned result is reported over three to five seeds with a statistical test. Finally, all timings come from one laptop CPU and are bounds rather than automotive measurements.

16. Reproducibility

All code, configurations, evaluation harness and result summaries are available at <https://github.com/shishirbit/can-ids-evaluation>. The repository is organised as follows. Dataset preparation and download live in `scripts/`; the detectors in `src/galaxy_its/models/`; the identifier-agnostic representations in `relative.py` and `residual.py` under `src/galaxy_its/data/`; the evaluation harness in `operating.py` under `src/galaxy_its/eval/`. The experiment drivers are `train_sttf.py`, `baseline_frames.py`, `id_agnostic.py`, `cross_vehicle.py`, `per_vehicle_calib.py` and `pair_detector.py`; statistical comparison is in `compare_models.py` and the figures are produced by `make_figures.py` and `make_figures2.py`. Every number quoted in this paper is recorded in `experiments/paper_numbers.json` together with the script that produced it, and the figures are generated from that file so that text, tables and plots cannot diverge.

The raw corpora are not redistributed: Car-Hacking and ROAD are obtained from their authors and can-train-and-test from its repository, using the download scripts provided. A full replication requires approximately 11 GB of disk for the corpora, a CUDA-capable GPU with 8 GB of memory, and about 12 hours of compute for all experiments reported here; individual experiments take 3–20 minutes. Random seeds, hyperparameters and split definitions are fixed in the experiment drivers and recorded in each run’s `results.json`.

17. Conclusion and future work

We have argued, and measured, that the quantities deciding whether a CAN intrusion detector can be deployed are not benchmark accuracies but the number of attack events caught per hour of false alarms and the behaviour of the detector on a vehicle it has not seen. Under an event-level protocol with an explicit false-alarm budget and an alarm duty cap, the most widely used benchmark turns out to reward learning an injection timetable; a 98 k-parameter model matches a 1.05 M-parameter one on the hardest public corpus, both stopping at a ceiling imposed by the data; five evaluation choices each reverse model rankings; and identifier-embedding detectors, the standard design, detect nothing on an unseen vehicle at a usable alarm rate. Expressing each frame relative to its own arbitration identifier restores 69–78% of that detection without any labels from the target vehicle, and running two such representations as separate detectors with OR-ed alarms reaches 81% at 0.3/h false alarms for 0.20 MB of parameters and roughly 40% of one CPU core.

Future work follows directly from what remains hard. First, masquerade and other contextual attacks need explicit models of cross-identifier signal consistency; the tier analysis here gives a concrete target and a way to measure progress. Second, the protocol should be extended to suppression attacks, which remove frames rather than adding them and therefore need a different notion of episode. Third, the cross-vehicle study should be repeated on a fleet-scale corpus with more than four vehicles, ideally including vehicles from a third manufacturer. Finally, we recommend that future CAN IDS results be reported at a stated false-alarm budget, with an alarm duty cap, on whole held-out drives, and – wherever the data permit – on a vehicle the detector has never seen.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

First Author: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft. **Second Author:** Validation, Formal analysis, Writing – review & editing. **Third Author:** Supervision, Project administration, Writing – review & editing.

Data availability

The three corpora used are public and are obtained from their original providers; the code and result summaries are available at <https://github.com/shishirbit/can-ids-evaluation>.

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